CS-405 Deep Learning **Quiz-03: Generative Adversarial Networks (GANs)**

Name: ____________________________ **Reg No:** ____________________________

Total Marks: 10

1. (3 points) (a) (2 points) What are the two networks in a GAN and what are their respective roles?

   (b) (1 point) Name two major problems that can occur during GAN training.

2. (4 points) In a Wasserstein GAN (WGAN), the critic (discriminator) outputs the following scores:

$$C(x_{real}) = 2.5, \quad C(G(z)) = -1.2$$

   (a) (2 points) Calculate the WGAN loss for this batch.

   (b) (2 points) The WGAN uses a gradient penalty with coefficient $\lambda = 10$. If the gradient norm $\|\nabla_{\hat{x}} D(\hat{x})\|_2 = 1.5$, calculate the gradient penalty term.

3. (3 points) State whether the following statements are **True** or **False**:

   (a) In a standard GAN, the discriminator outputs a probability between 0 and 1.

   (b) Mode collapse occurs when the generator produces diverse and varied outputs.

   (c) Conditional GANs (cGANs) add condition information to both the generator and discriminator.

Answer Key

1(a) Generator (G): Creates fake data from random noise to fool the discriminator. Discriminator (D): Distinguishes between real and fake data.

1(b) Mode collapse, vanishing gradients, non-convergence/oscillations (any two).

2(a) $\log(1 - 0.3) = \log(0.7) = -0.357$

2(b) $-\log(0.3) = -(-1.204) = 1.204$

2(c) $\log(1 - D(G(z))) = \log(0.7) = -0.357$ (same as 2a)

3(a) WGAN loss $= \mathbb{E}[f_w(x_{real})] - \mathbb{E}[f_w(G(z))] = 2.5 - (-1.2) = 3.7$

3(b) Gradient penalty $= \lambda \cdot (\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2 = 10 \times (1.5 - 1)^2 = 10 \times 0.25 = 2.5$

4(a) True

4(b) False (mode collapse means generator produces identical or very similar outputs)

4(c) False (WGAN critic uses linear/no activation, not sigmoid)

4(d) True